Not All Spines Are Created Equal: How CT Segmenters Fail on Lumbosacral Transitional Vertebrae

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Abstract

Wrong-level spine surgery remains a persistent never-event, and the lumbosacral transitional vertebra (LSTV, 4–30% prevalence) is a recognized anatomic cause. We audit TotalSegmentator 2.4.0—a widely used CT segmentation system, available in 3D Slicer—and show that on lumbarization cases its output reproduces the pattern of a one-level miscount: its `total` task lacks an L6 class and shifts lumbar labels caudally by one level, and junction-DSC drops 29 points (95% CI 10–47) on Any-LSTV ($0.824 \rightarrow 0.534$; n=7 cases with a scorable junction). No existing CT benchmark surfaces this—not TotalSegmentator's own evaluation, not Li et al. [2025] (the per-class DSC frontier on the matched COLONOG split), and not VERIDAH (vertebra-*labeling* SOTA on a private cohort, on pre-localized crops with no pelvis). We contribute: (i) an LSTV-stratified evaluation protocol (per-class DSC, junction-DSC over a 40 mm L5/S1 window, per-class voxel confusion at the junction) that measures the level shift and frames it as a plausible automated wrong-level-risk mechanism; (ii) CTSPINOPELVIC1K—1,153 CT records across 802 patients with unified spinopelvic masks, 33 LSTV-positive patients in a six-way phenotype taxonomy derived from the source annotations' filenames and lumbar counts, and Castellvi typing (I–IV) of all 33 by a two-resident consensus (κ=0.74, no senior read); (iii) the first publicly released merge-based dual-spinopelvic CT segmenter, exceeding TotalSegmentator on sacrum ($0.817 \rightarrow 0.969$) with released weights; (iv) a same-data unmerged control with an explicit L6 class that still labels

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most ground-truth L6 voxels L5, indicating that the training-side L5/L6 merge, not the added class, removes the collision, while a residual L4/last_lumbar collision on the one held-out count-style sacralization case motivates a downstream sequence predictor. Only 7 held-out records carry an LSTV subtype, in a split drawn by record rather than patient; every stratified number is reported with its n.

1 Introduction

Wrong-level spine surgery is a persistent never-event in operative spine care, and miscounting vertebral levels in the presence of an LSTV is a recognized cause [Konin and Walz, 2010, Carrino et al., 2011]. The mechanism is well documented: a surgeon counts lumbar bodies upward from the sacrum, finds five (the sixth goes unrecognized), and labels the bottom body L5 when it is anatomically L6. Surgery proceeds one level caudal to the intended target.

A widely used CT segmentation system is TotalSegmentator [Wasserthal et al., 2023]: an nnU-Net trained on 1,204 routine CTs to delineate 104 anatomic structures and available as a 3D Slicer extension. The original paper reports an aggregate test-set Dice of 0.943 across all 104 classes and qualitatively documents two spine failure modes affecting neighboring-vertebrae label confusion (1% of cases) and neighboring-rib confusion (12%). TotalSegmentator is neither the only public CT spine segmenter nor the per-class DSC frontier on the matched CTSpine1K + CTPelvic1K (COLONOG) split (§2); it is, however, a widely used system whose output downstream tools can consume. This paper concerns one anatomic class on which TotalSegmentator does not perform at its headline level.

The risk mechanism. On lumbarized spines, TotalSegmentator produces an output pattern that matches the mechanism of the surgical miscount. Its label vocabulary (the `total` task of version 2.4.0, audited here) contains L1–L5 but no L6 class. On a lumbarized spine (six lumbar bodies plus sacrum), the model distributes its five lumbar labels across the six bodies by shifting caudally by one anatomic level: anatomical L6 receives the L5 label, anatomical L5 the L4 label, and so on. A plan that took the disc between predicted-L4 and predicted-L5 as L4–L5 would land on the anatomical L5–L6 disc. We present this as a *plausible automated wrong-level-risk mechanism*: the paper measures how often the shifted output occurs and what it looks like at the junction. It does not evaluate surgical-planning software, reader behaviour or downstream workflow, and no wrong-level event is attributed to a segmenter here.

The architectural alternative: VERIDAH. Möller et al. [2026] label transitional vertebrae with a per-vertebra classifier trained with L5/L6 merged, followed by a constrained sequence predictor that restores the count (§2). Whether a voxel-wise network trained at full resolution with the same merge needs the sequence predictor at all has never been tested; we supply that test.

The evaluation gap. Neither TotalSegmentator, VerSe [Sekuboyina et al., 2021], CTSpine1K [Deng et al., 2021], nor CTPelvic1K [Liu et al., 2021] report LSTV-stratified performance; earlier CT labelers excluded the failure mode [Sekuboyina et al., 2020], and VERIDAH reports anomaly performance on a private cohort. A production model can therefore emit a locally well-formed, caudally shifted prediction on a 4–30%-prevalence variant with no error signal at evaluation time.

Contributions.

(i) **The level shift in TS, quantified.** An LSTV-stratified evaluation protocol (per-class DSC, junction-DSC over a 40 mm L5/S1 window, per-class voxel confusion at the junction) measures TS's systematic one-level caudal shift on lumbarization (L4=0.388, L5=0.291 over 14 patients; 95–97% of GT-L6 junction voxels labeled L5 in 2 of the 3 lumbarization patients with a scorable junction, §3) and the structural absence of L6 across all evaluation cases, and frames the shift as a plausible automated wrong-level-risk mechanism.

(ii) **CTSPINOPELVIC1K: a stratified spinopelvic benchmark.** 1,153 abdominal CT records across 802 patients, unified spinopelvic masks, 33 LSTV-positive patients in a six-way subtype taxonomy (lumb n=14; sacr_count n=9; full sacralization n=6; semi-sacralization n=2; ambiguous n=2), and a registration-free fusion pipeline that places each source mask on the acquisition it was drawn on, with its orientation corrections disclosed (Appendix G).

The subtypes come from the source annotations (filename qualifiers and lumbar counts), not from image reads; the Castellvi layer (I–IV, all 33) is a two-resident read made after peer review, with κ=0.74 and no senior adjudication (§5). Both source datasets annotate the COLONOG cohort, but neither encodes pelvic context with LSTV stratification.

(iii) **First publicly released merge-based LSTV-handling dual-spinopelvic CT segmenter.** A 5-fold CV nnU-Net 3d_fullres baseline on H200 hardware adopting VERIDAH's training-side merge (last_lumbar = L5+L6) at the voxel level, with pelvic-class supervision in a single forward pass. We exceed TS on sacrum and uniquely deliver both spine and pelvis with released weights (Table 4); §2 details the reproducibility tiers separating openly-released from paper-only baselines.

(iv) **An unmerged control: the merge is necessary and insufficient.** A same-data, same-architecture control trained *without* the merge, with an explicit L6 class and L6 oversampling, still labels most GT-L6 voxels L5 (§8.1), indicating that the merge, not the added class, removes the L5/L6 collision driving TS's level shift. The merge in turn exposes a residual L4/last_lumbar collision on count-style sacralization (one held-out case), pointing to the component of VERIDAH's success attributable to its sequence predictor—a direction VERIDAH's authors could not ablate.

2 Related Work

Wrong-level spine surgery and LSTV. [Konin and Walz, 2010, Carrino et al., 2011, Sekharappa et al., 2014] document LSTV-induced miscounting (4–30% prevalence) without addressing automated segmentation as a miscount-propagation pathway. Kwak et al. [2025] detect LSTV on plain radiographs (binary; different modality).

Production CT spine and pelvic segmentation. TotalSegmentator [Wasserthal et al., 2023], an nnU-Net trained on 1,204 CTs, has no L6 class in its audited `total` task (2.4.0) and no LSTV stratification; we audit it as a widely used production system, not as a per-class DSC frontier. Li et al. [2025] hold the published per-class DSC frontier on a COLONOG split (731 scans; 440/146/145) but release neither weights nor code (paper-only) and do not stratify by LSTV. Liu et al. [2025] release FracSegNet (PENGWIN+CTPelvic1K), a pelvic-only cascaded 3D nnU-Net providing sacrum and L/R hipbone; their qualitative note of sacralization degradation (their §4.3) corroborates the LSTV gap our benchmark surfaces. Table 4 compares all baselines, Appendix H CT collections.

Vertebra-labeling SOTA: VERIDAH. Möller et al. [2026] address transitional-vertebra *labeling* (distinct from segmentation): a per-vertebra classifier trained with L5/L6 (and T12/T13) merged, plus a constrained sequence predictor enforcing 4–6 lumbar bodies (Appendix F). On a private CT cohort of 1,536 volumes VERIDAH reports 99.18% subject-level perfect labeling versus prior CT SOTA [Meng et al., 2022] 77.81%. It requires vertebra locations from an upstream instance segmentation (for MRI, SPINEPS [Möller et al., 2024]), excludes the pelvis, and validates privately. Its ablation (their Table 5) shows the sequence predictor adds substantial value, but the comparison against direct semantic segmentation on a public LSTV-stratified split is missing—the gap our baseline (§7) fills.

Benchmarks and evaluation rigor. VerSe [Sekuboyina et al., 2021] (374 CT scans) and CT-Spine1K both label L6 (label 25), but neither reports LSTV-stratified results and neither has pelvic GT; earlier CT labelers explicitly excluded the failure mode [Sekuboyina et al., 2020]. Stratified safety-critical reporting [Maier-Hein et al., 2024] motivates §6.

3 Failure-Mode Analysis: TS on LSTV

We evaluate TotalSegmentator 2.4.0 (`total` task; TS means this version and task throughout) zero-shot at full resolution on the full CTSPINOPELVIC1K matched cohort. TotalSegmentator was not trained on any CTSPINOPELVIC1K case, so every case is valid; this provides 1,152 of CTSPINOPELVIC1K's 1,153 records (one record dropped due to TS inference failure) aggregated to 801 patients via patient-level merging, including 33 LSTV-positive patients. Full-resolution mode isolates the failure as a property of label vocabulary and training distribution rather than inference

Table 1: TS zero-shot stratified by LSTV phenotype on CTSPINOPELVIC1K, full-resolution mode; n is patients (per class, with 95% intervals, in Appendix E). Junction DSC over a 40 mm L5/S1 axial window is scored on fused records (n_{jxn}; two count-style records lack GT-L5 there, §3). †: no scorable junction. *One patient.

Subgroup	n	**L1**	**L2**	**L3**	**L4**	**L5**	**Sac**	**Hips**	**Jxn DSC**	n_{jxn}
All cases	801	0.911	0.906	0.913	0.914	0.913	0.817	0.929	0.818	341
Normal	768	0.921	0.918	0.929	0.936	0.938	0.817	0.929	0.824	334
Any LSTV	33	0.681	0.615	0.523	0.367	0.314	0.822	0.929	0.534	7
Lumbarization	14	0.878	0.853	0.690	0.388	0.291	0.833	0.928	**0.357**	3
Sacralization (count-style)	9	0.265	0.100	0.055	0.001	0.000	0.814	0.926	0.697	2
Semi-sacralization	2	0.951*	0.951*	0.951*	0.955*	0.956*	0.824	0.937	†	0
Sacralization (full)	6	0.718	0.676	0.673	0.724	0.793	0.826	0.932	0.636	2
Ambiguous	2	0.952	0.951	0.882	0.678	0.374	0.771	0.932	†	0
L6 EMIT										0 / 1152

resolution. TotalSegmentator 2.15.0 (1 July 2026) added a `total_v3` task with an L6 class, after this benchmark ran (30 April); it is not evaluated here.

Table 1 reports performance by phenotype. On normal anatomy (n=768) TotalSegmentator performs at its published level. On Any-LSTV (n=33) every lumbar metric drops; lumbarization collapses caudally (L1=0.878 → L4=0.388 [0.18, 0.60] → L5=0.291 [0.10, 0.51] over 14 patients; case-bootstrap 95% intervals, Appendix E). Junction-DSC is scored on fused records, whose ground truth holds spine and sacrum: 334 normal patients and 7 LSTV patients (3 lumbarization: 0.22, 0.23, 0.62; 2 count-style sacralization: 0.47, 0.93; 2 full sacralization: 0.43, 0.84), so the 29-point Any-LSTV drop in the abstract (95% CI 10–47) rests on 7 patients. In the two count-style records the window holds no GT-L5 (0.93 is sacrum alone); requiring it leaves 5 LSTV patients at 0.469, a 35.5-point drop. In two of the three lumbarization junctions TS labels 95% and 97% of GT-L6 voxels L5; in the third it leaves them background. Per patient, L5 DSC falls below 0.5 in 10 of 14 lumbarization cases (Wilson 95% interval 45–88%) and in 9 of 9 count-style sacralization cases (70–100%), against 9 of 750 normal cases (0.6–2.3%). L6 predictions are structurally zero across all 1,152 records (0–0.3%).

The level-shift mechanism. The asymmetry between lumbarization (severe, caudally progressive) and count-style sacralization (uniform collapse) follows from a single fact: the output head of TotalSegmentator's `total` task has no L6 class. For 6-vertebra anatomies the model compresses six bodies into five labels: the most rostral remains L1, each successive body is misaligned, and the last, anatomically L6, is labeled L5. For 4-vertebra anatomies it emits standard L1–L5, overshooting the GT count by a uniform $+1$ level. Both are segmentation analogs of the surgical miscount [Konin and Walz, 2010].

4 Case Study: the Level Shift at the Junction

Aggregate metrics establish the population pattern; the case study shows the mechanism on one patient's output. Token 215 is a lumbarization case (six lumbar bodies; Castellvi Ib by consensus, one of the six cases the two readers graded differently, §5). Table 2 reports per-class DSC: L1 and L2 are well-placed (0.954, 0.927), L3 is partially correct (0.594), and **L4, L5, and L6 all collapse to dice zero**—no GT-L4 or GT-L5 voxel correctly labeled and no L6 voxel emitted. Pelvic classes stay near Normal: the failure is local to the lumbosacral count.

Table 2: Token 215 (lumbarization, 6 lumbar bodies): per-class TS DSC. L4, L5 and L6 collapse to zero because TS compresses 6 bodies into 5 labels; pelvic classes are unaffected. † TS has no L6 label.

Class	L1	L2	L3	L4	L5	L6	Sacrum	Hip$_L$	Hip$_R$
DSC	0.954	0.927	0.594	**0.000**	**0.000**	**0.000**†	0.817	0.922	0.926

Why the pattern matters. The output is locally well-formed: five compact vertebral predictions in roughly correct positions, the shift detectable only by counting bodies (six lumbar, not five), which

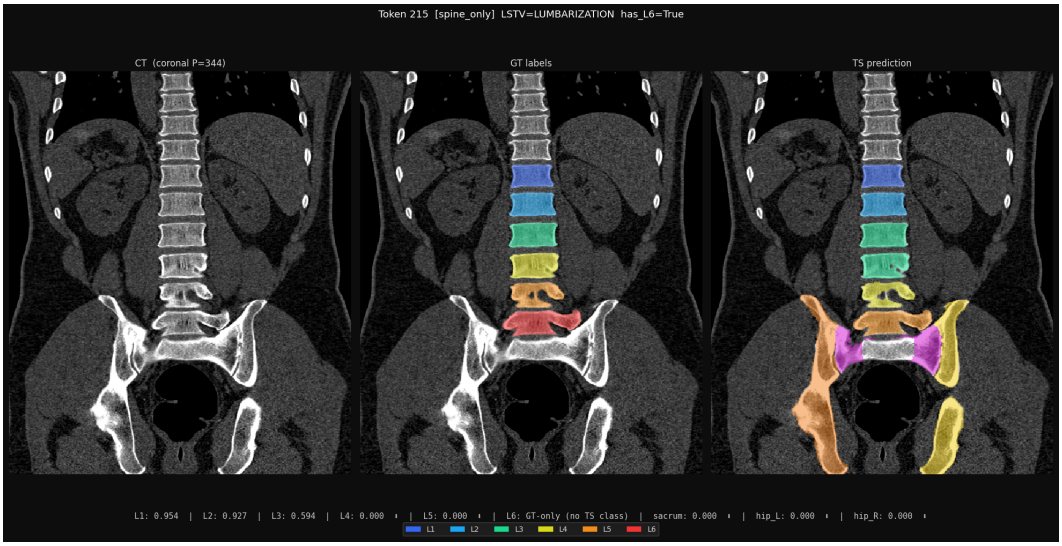


Figure 1: Token 215, lumbarization (6 lumbar bodies): TS predictions on CT. Predicted-L3 absorbs anatomical L3 and L4, predicted-L4 sits on anatomical L5, predicted-L5 on anatomical L6, which has no label in TS's output head. Five compact, well-formed predictions, every label below L2 one level off. Quantified in Table 2.

the output itself does not flag. Whether a surgeon or a planning tool would act on such an output is not tested here; the claim is the mechanism and its frequency, not a measured clinical harm.

5 CTSpinoPelvic1K: Construction and Composition

CTSpinoPelvic1K fuses CT scans from COLONOG [Smith et al., 2015] with annotations from CTSpine1K [Deng et al., 2021] (per-vertebra masks in the VerSe label convention) and CTPelvic1K [Liu et al., 2021] (sacrum and bilateral hips). CTSpinoPelvic1K's contribution is the patient-level fusion of CTSpine1K's vertebra masks with CTPelvic1K's pelvic masks plus the LSTV-stratification overlay neither source provides. Of the 802 matched patients, 342 have one record with fused (spine + pelvis) ground truth, 351 are separate-mode (a spine record and a pelvic record on different acquisitions), 89 are spine-only and 20 pelvic-only: 1,153 records, 53 of them from the 33 LSTV patients (Appendix A, Table 7).

Release used. Every segmentation number in this paper was computed on the April 2026 release (the HuggingFace export of 25 April 2026; training began 3 May): 1,153 records across 802 patients and a ten-class scheme (background, L1–L6, sacrum, left and right hip) merged to nine classes for training. Its files changed after 25 April: on 27 April, CTs that had skipped resampling into their mask's grid were rewritten and, after a TotalSegmentator laterality audit, six patients' hip labels were swapped left for right; on 1 May, the unannotated voxels of the 811 single-annotation records became an ignore label and the manifests were redrawn (Appendix G). Neither these records nor the evaluated split is yet at a public location this paper cites (Appendix A). The Castellvi grades and κ are instead the two-reader reads completed on 4 September 2026 (§5), released from Zenodo v8 onward. The archive of record is the Zenodo concept DOI 10.5281/zenodo.22139642, whose current version is v11 (10.5281/zenodo.22797014, 18 September 2026); it keeps the same 802 patients but changes the records and the label scheme (Appendix A). Nothing reported here was recomputed on v11.

Patient-anchored fusion and orientation review. Patient identity is resolved via canonical UID matching (825/825 patients). Each mask is assigned to its source CT acquisition by intra-patient bone-HU maximization, $s^*(m) = \arg\max_{s \in \mathcal{S}_p} |\{v \in \mathcal{L}(m,s) : I_s(v) > 200\,\text{HU}\}|/|\mathcal{L}(m,s)|$, a metadata-free oracle since `SeriesDescription` fields are unreliable in TCIA releases (99.6% spine, 100% pelvic placement). A review of placement renders found nine patients whose CT and masks looked front-to-back inverted; each was mirrored front to back. A mirror is not a rotation: the

two rendered since (tokens 75, 480) show the liver on the left, so they are left–right mirror images, and 480's hip-label swap matched its labels to the mirrored image, not the patient (Appendix G). We adopt VERIDAH's training-side merged-label scheme at the voxel level: 0 = bg, 1–4 = L1–L4, **5 = last_lumbar** (anatomical L5 in 5-vertebra spines; both L5 and L6 in 6-vertebra spines), 6 = sacrum, 7–8 = left/right hip, 9 = ignore.

Six-way LSTV taxonomy, and where its labels come from. No subtype in this benchmark was assigned by reading an image. Each is a deterministic function of the two source annotations: the number of lumbar labels in the CTSpine1K mask (a radiologist's count, in VerSe convention, so a sixth lumbar body carries VerSe label 25) and the qualifier tag in the CTPelvic1K mask filename (`sacralization`, `semisacralization`), whose origin CTPelvic1K does not document. The six subtypes are **normal**, **lumb** (six lumbar labels or label 25 present), **sacr_count** (four lumbar labels and no qualifier), **semisacralization** and **sacralization** (qualifier tag; the lumbar count is five), and **ambiguous** (the two sources disagree). Two of the six therefore rest entirely on a filename tag, and the count subtypes inherit each source's naming convention for the transitional segment. An earlier substring-match parser collapsed semisacralization into full sacralization, merging two phenotypes the source distinguishes; we identify and fix this. Under TS (Table 1), full sacralization is near-normal in 3 of its 5 spine-labelled patients (one intermediate, one collapsed); semi-sacralization has one spine-labelled patient, near-normal, and no scorable junction.

Partial-annotation training protocol. Training only on fused records discards 70% of the data. We use nnU-Net v2's `ignore_label` [Isensee et al., 2024] so that partial records train too: voxels in unsupervised regions are written as label 9, contributing zero to Dice and cross-entropy. Of the 1,153 records, 782 carry spine labels (classes 1–5) and 713 pelvic labels (6–8), against 342 fused records. Training tags follow the supervised anatomy: a pelvic-only record of a lumbarization or count-style patient, or a spine-only record of a (semi-)sacralization patient, is tagged normal.

Castellvi annotation layer. The 6-way taxonomy captures the *count* dimension of LSTV. The Castellvi classification [Castellvi et al., 1984] captures the orthogonal *morphological* axis: Type I (dysplastic transverse process), II (pseudoarticulation), III (bony fusion), or IV (mixed II/III), each `a` (unilateral) or `b` (bilateral). It is what a CT segmenter would need to predict to flag LSTV for review rather than silently emit a caudally shifted label. Two radiology residents, both co-authors, each graded all 33 LSTV cases after peer review, scoring the bony junction per side in every variant; the reads were complete by 4 September 2026, and reader 2's grades for five cases are second reads recorded on 23 August. The reviewed version's grades came instead from a one-column sheet of May 2026, described there as a joint two-resident consensus, so the readers were not naive to the cases. They agreed on 27 of 33 (Cohen's κ=0.74 on the full grade with laterality, 95% interval 0.55–0.92; κ=0.70 on type I–IV alone, 0.48–0.92; Fleiss–Cohen–Everitt standard errors; n=33). Five of the six disagreements cross the II/III boundary (articulation versus bony fusion) and one the I/II boundary; the consensus kept reader 1's grade in all six. Against the reviewed table, five grades changed (four III→II, one IIIb→Ib): Type I rose from 1 to 2 of 33, IIIb fell from 19 to 16, and three of the six full sacralizations are now Type II (Appendix B). There was no attending or neuroradiologist read, no blinded re-read and no screening of the `normal` cohort, so the grades are a two-trainee consensus and every Castellvi-stratified statement below inherits that limit. Both reads and the consensus are released with the data from Zenodo v8 on.

Table 3: Two-reader consensus Castellvi grade (September 2026) for the 33 LSTV patients, by count subtype. Pool: patients with an LSTV-tagged record in the 979-record training pool; Test: patients with an LSTV-tagged held-out record. Token 4 is in both; five patients have records on both sides (§7).

Subtype	**Ib**	**IIa**	**IIb**	**IIIa**	**IIIb**	**IV**	**Total**	**Pool**	**Test**
Lumbarization	2	1	3	1	6	1	14	11	3
Sacralization (count-style)	0	0	0	0	9	0	9	8	1
Sacralization (full)	0	1	2	0	1	2	6	4	2
Semi-sacralization	0	2	0	0	0	0	2	2	0
Ambiguous	0	0	1	0	0	1	2	2	1
Total	2	4	6	1	16	4	33	27	7

Morphology-stratified failure interpretation. The cross-tabulation (Table 3) surfaces two findings with direct bearing on model behavior. (1) **All 9 count-style sacralization cases are Castellvi IIIb**—bilateral bony fusion, a grade both readers assigned to all nine—and 16/33 (48%) of LSTV overall is IIIb. On the one held-out count-style case (n=1), the residual L4/last_lumbar collision (§8.1) therefore occurs where the discriminating cue is locally observable: bony fusion of the bottom lumbar to sacrum is visually distinct from a normal L5–S1 junction, yet the segmenter still emits last_lumbar for those voxels. On that case, direct semantic segmentation did not use the Castellvi correlate of the count anomaly although it is morphologically present in the input—an argument, from one case, for downstream Castellvi-aware reasoning at the junction (VERIDAH-style global count constraints or a dedicated Type III morphology head). (2) Only 2/33 cases are Type I. Selection is by lumbar count and filename tag, so an LSTV with five lumbar bodies and no tag, as a Type I dysplastic transverse process can be, is not in the cohort; this share is not a prevalence. Castellvi-stratified statements are therefore scoped to the conspicuous regime (Castellvi II–IV, 31 of 33 cases), and Type I is uncharacterized here. The two Type I patients (tokens 175 and 215, both Ib lumbarizations; 215 is the case study) are in Table 1's lumbarization and Any-LSTV rows (without them TS's lumbarization L4/L5 DSC is 0.452/0.339, n=12) and in fold 4 of the unmerged control (§8.1); neither is held out.

6 Evaluation Protocol

We define four metrics as the minimum required to attribute LSTV failure modes to the wrong-level-risk mechanism [Maier-Hein et al., 2024]: (1) per-class DSC stratified by 6-way LSTV subgroup, exposing class-level degradation; (2) junction-DSC over a 40 mm L5/S1 axial window, where the level shift lands; (3) per-class voxel confusion at the L5/S1 junction, which exposes structured failure modes (the level shift) that DSC alone cannot characterize; and (4) for the merged-label baseline, three confusion blocks on the L4/last_lumbar boundary that drives the residual collision (Table 6) plus last_lumbar specificity on sacr_count. Every stratified number carries its n; single-record cells are marked, TS means over four or more patients carry case-bootstrap 95% intervals and our three-record lumbarization stratum is given per case (Appendix E), and proportions carry Wilson 95% intervals.

7 Spinopelvic Baseline

Architecture and training. nnU-Net v2 [Isensee et al., 2020, 2024], `3d_fullres`, with the ResEnc-M planner at a 100 GB GPU budget (`nnUNetResEncUNetPlans_100G`). Spacing $0.80\times0.75\times0.75$ mm, patch $\mathbf{256}\times\mathbf{320}\times\mathbf{320}$, batch 2; 7-stage residual encoder with deep supervision and a 9-channel output (Appendix D); ignore-label masked from loss. Class-reweighted DC_and_CE loss ($\mathbf{w}$=$[0.5, 1, 1, 1, 1, \mathbf{1.5}, \mathbf{2.0}, 1, 1]$, boosting last_lumbar and sacrum); SGD Nesterov 0.99, λ=3×10^{-5}, PolyLR 10^{-2}, 500 epochs (250 train + 50 val iters); 5-fold CV, grouped by patient and stratified by LSTV subtype, over a 979-record pool (the release's 805-record train and 174-record validation manifests); the 174-record test manifest is held out.

The held-out split is drawn by record. The test manifest was stratified by LSTV class and acquisition mode but drawn per record, not per patient: 92 of its 167 patients are separate-mode patients whose other acquisition, labelled only for the other region, is in the pool and trained four of the five fold models; among LSTV patients these are 401, 555, 123 and 4, and patient 104, whose only held-out record is pelvic and scored as Normal (as is a pelvic record of patient 672). Every junction-DSC value comes from a fused record, which has no sibling (Appendix A).

Two-tier LSTV emphasis. **Queue-level oversampling**: all five non-normal subtypes duplicated to 25% of per-fold training samples (matching VERIDAH's 4× rate). **Class-reweighted CE** (above). No per-patch foreground biasing, which would confound the residual-collision measurement.

Hardware. H200 NVL (140 GB HBM3e), 24 CPU workers and a local-NVMe cache (Appendix D); per-epoch $\approx$ 400–700 s, per-fold $\approx$ 3 days for 500 epochs. Reducing the GPU memory budget forces either smaller patch (loses spinopelvic context) or coarser spacing (loses Castellvi-defining detail); 100 GB is the smallest budget fitting both.

Table 4: Per-class DSC ($\times 100$). **last_lumbar** merges L5+L6 (L5 on 5-vertebra anatomy); Hip_m merges LH+RH. TotalSegmentator: Normal subgroup zero-shot (Table 1), Hips as Hip_m. Top block: from those works' own test sets. Bold: column best. Ours: mean of five fold checkpoints; n counts records. †Held-out LSTV-tagged records with GT for the class (5 for L1–L4, else 4).

Method	L1	L2	L3	L4	L5	last_lumbar	Sacrum	LH	RH	Hip_m
Reproduced from prior work; not evaluated on CTSPINOPELVIC1K *(Li et al. on COLONOG split, FracSegNet on PENGWIN test set)*										
3D U-Net	77.19	78.34	81.71	82.86	92.13	—	90.02	93.74	93.42	—
ResNet-50	80.29	76.41	80.62	83.77	91.05	—	88.37	94.72	94.01	—
ResNet-101	74.57	78.11	78.54	83.84	86.35	—	89.05	95.69	94.42	—
SwinUNETR	88.87	83.12	84.36	89.04	92.15	—	91.38	92.33	93.38	—
AttentionUNet	80.26	75.84	79.64	82.32	90.39	—	92.48	93.41	95.45	—
nnU-Net [Isensee et al., 2020]	89.35	91.37	92.41	93.37	95.74	—	96.72	95.50	96.78	—
Li et al. (CFFM+LT) [Li et al., 2025]	92.98	92.36	93.92	93.31	**97.07**	—	96.98	94.37	96.77	—
FracSegNet [Liu et al., 2025]	—	—	—	—	—	—	**98.7**	**99.3**	**99.2**	—
Reproducible baseline (publicly-released weights), evaluated zero-shot on CTSPINOPELVIC1K										
TotalSegmentator [Wasserthal et al., 2023]	92.1	91.8	92.9	93.6	93.8	—	81.7	—	—	92.9
Ours: CTSPINOPELVIC1K*, five fold checkpoints on the held-out split, partial-annotation supervision*										
Ours, Normal (n=167)	**95.3**	**95.5**	**95.9**	**96.7**	—	**97.1**	97.1	95.0	93.6	**98.8**
Ours, Any-LSTV (n=4–5)†	72.1	55.8	37.3	17.7	—	66.9	90.7	98.3	98.3	98.4

Table 5: Five fold checkpoints on the 174 held-out records, mean ± std across checkpoints; n counts records (Normal: 108 with spine GT, 109 pelvic). LL EMIT: records with ≥ 100 last_lumbar voxels. Dashes: no GT. Δ: paired against TS zero-shot on the same records (its Hip_m: mean of per-side DSC). *n=1: ± omitted. ‡One record scorable; ± is five checkpoints on it.

Subgroup	n	L1	L2	L3	L4	last_lumbar	Sac	Hip_m	Jxn DSC	LL emit
Normal	167	.953±.008	.955±.005	.959±.004	.967±.005	.971±.005	.971±.002	.988±.001	.907±.016 (n=50)	167/167
Lumbarization	3	.872±.050	.611±.123	.377±.147	.213±.114	**.698±.038**	.946±.004‡	.960±.002‡	**.946±.007**‡	3/3
Sacralization (count)*	1	.988	.959	.734	.187	—	—	—	—	1/1
Sacralization (full)	2	.001±.002‡	.000±.000‡	.001±.002‡	.062±.083‡	.583±.136‡	.910±.039	.989±.000	.927±.068‡	2/2
Ambiguous*	1	—	—	—	—	—	.862	1.000	—	1/1
Δ vs. TS (lumb)	—	−8.0	−32.2	−32.0	−3.8	N/A	+13.6	+5.1	**+72.5**	—
Δ vs. TS (sacr_count)	—	+98.8	+95.9	+73.3	+18.6	—	—	—	—	—

8 Results

Headline, with its n. On the *one* junction-scorable held-out lumbarization case (token 167, a fused record whose ground truth holds both L5 and sacrum), the merged-label baseline closes the junction-DSC gap: it rises from TS = 0.221 to ours = **0.946** ± 0.007 (mean ± std of five fold checkpoints; ensemble 0.950), a paired Δ= **+72.5** ± 0.7 points (+58.9 against TS's unpaired mean of 0.357 over its three junction-scorable lumbarization patients). The ± is the spread of five checkpoints on this one patient, not a population spread. The model emits non-degenerate last_lumbar on all 3/3 held-out lumbarization cases (Wilson 95% interval 44–100%) with per-case ensemble last_lumbar DSC of 0.645, 0.666 and 0.725 (5-fold mean 0.698 ± 0.038), eliminating the structural absence that drives TS's level shift. A residual class collision persists on count-style sacralization: on the single held-out sacr_count case, evaluated by all five fold checkpoints (∼720K GT-L4 voxel evaluations), **82.7%** of GT-L4 voxels are mis-predicted as last_lumbar (§8.1, Block 2), with per-case L4 DSC of 0.187 ± 0.337 across folds (one checkpoint scores 0.79, the others below 0.10; ensemble 0.032). On pelvic classes, our model exceeds TotalSegmentator on sacrum by Δ= **+15.2** ± 0.2 points, unpaired (TS = 0.817 over the 712 patients with sacral GT → **0.969** ± 0.002 over the 113 held-out records with sacral GT; 0.970 ensemble; TS on those 113 records: 0.815) and establishes state-of-the-art for openly-released dual spinopelvic CT segmentation (Table 4).

LSTV-stratified per-subgroup view. Table 5 exposes the LSTV-induced degradation that aggregate per-class DSC masks; the residual L4 DSC drop on the one held-out sacr_count case (§8.1) persists. Semi-sacralization has no held-out case and is therefore unevaluated, not measured. last_lumbar is emitted on all 4 held-out LSTV records with last_lumbar GT (Wilson 51–100%); on the sacr_count record emission is itself the error (Block 3), and two pelvic-only records cannot be judged.

Table 6: Residual class confusion under merged labels on the held-out split, focused on L4/last_lumbar. Voxel counts aggregated across 5 fold checkpoints; percentages from summed counts. Block 1: GT-last_lumbar on the 3 lumb cases ($\sim$4.06M voxels). Block 2: GT-L4 on the 1 sacr_count case ($\sim$720K). Block 3: pred-last_lumbar on that case ($\sim$595K).

Block	last_lumbar	sacrum	L4	background
Block 1: last_lumbar GT on lumb (n=3)	**58.2%**	0.4%	40.1%	1.2%
Block 2: L4 GT on sacr_count (n=1)	*82.7%*	0.3%	**14.9%**	2.2%
Block 3: pred-last_lumbar on sacr_count (n=1)	—	0.0%	**100.0%**	0.0%

8.1 The Residual Class Collision

The merged scheme eliminates the L5/L6 collision driving TS's level shift. Each fold's checkpoint evaluates the same 174-case test split; voxel counts below pool 5 evaluations per case (15 case-fold pairs on the 3 lumb cases, 5 on the single sacr_count case). Across 5-fold predictions on lumb cases ($\sim$4.06M aggregated GT-last_lumbar voxels), **58.2%** of GT-last_lumbar voxels are correctly predicted as last_lumbar (Block 1), against TS's structurally-zero last_lumbar emission. The residual 40.1% misallocation to L4 is the cost of label-merging at the joint, not a re-emergence of TS's full collapse. The merge does, however, expose a residual collision at the L4/last_lumbar boundary on the sacr_count case: GT-L4 voxels are predicted as last_lumbar in **82.7%** of voxels (Block 2; $\sim$720K GT-L4 voxels aggregated across 5 folds), and predicted-last_lumbar voxels on that case are anatomically L4 in **100.0%** (Block 3). last_lumbar specificity on sacr_count = 0.778 ± 0.074 across folds. Voxels within a case are correlated, so these are dependent observations on one patient: a directional signal that motivates the sequence predictor, not a population estimate.

Same-data unmerged control. To separate the merge from the added class, we trained the same network (planner, patch, optimizer, 500 epochs, queue-level LSTV oversampling) on the same 1,153 records under the *unmerged* ten-class scheme (background, L1–L6, sacrum, hips, ignore), with an explicit L6 class weighted 4.0 in the cross-entropy and patch-level L6 oversampling on. Its folds come from a different record-level split with its own subtype map (count-style merged into sacralization, one more record tagged lumbarization); 149 of our 174 held-out records lie in its cross-validation pool, so it is read only inside its own folds. It did not learn L6. In the per-epoch validation confusion block, the share of ground-truth L6 voxels predicted as L5, averaged over the final 100 of 500 epochs, was 83%, 65%, 66% and 97% in folds 0, 2, 3 and 4, whose validation splits hold a lumbarization case (100%, 93%, 65% and 100% at the final epoch; fold 1 holds none), and the validation L6 Dice averaged 0.01–0.13 per fold over the same epochs against 0.91–0.95 for L5. The metric is computed on each epoch's validation patches and fluctuates (Appendix C); the fold-level means are the statistic. The 84.7–99.4% quoted during review was read from folds 0 and 4 about halfway through training, before folds 1–3 had started (Appendix C). With the same data and capacity, adding an L6 class did not remove the L5/L6 collision; merging does. This is a limit of dense segmentation at this LSTV sample size (12 lumbarization-tagged patients in its pool), not an impossibility.

Residual upper-lumbar caudal-shift cascade. The merge restores the bottom-most boundary (Block 1) but leaves a residual caudal shift in the upper lumbars on lumb cases (Table 5; two of the three cases shift, the third scores L1–L3 above 0.98). TS's shift sits at the bottom (no L6 class), ours through the upper lumbars, so on per-class L1–L4 TS is better on the same records, trading upper-lumbar precision for bottom-anchor correctness. Junction-DSC, centered on L5/S1, closes the gap there ($+72.5$ points, paired, on the one scorable case) regardless; the upper cascade needs a sequence predictor with count constraints (contribution iv): the merge is necessary but insufficient.

Interpretation. The model recovers the merged last_lumbar class on lumbarization, but on the one held-out count-style sacralization case the positional prior dominates in four of five checkpoints: the bottom-most lumbar visible in a patch is treated as last_lumbar regardless of the global count. The disambiguating signal is non-local; a $256\times320\times320$ patch does not reliably contain both the lumbosacral and the cervico-thoracic boundary needed to count. This is the case for VERIDAH's two-stage design: the merge addresses class imbalance; telling a 4-vertebra spine's last lumbar from a 5-vertebra spine's needs a component with global structure.

9 Discussion and Future Work

Generalization, hardware and future work. A spine segmenter without an L6 class has no correct label for a sixth lumbar body, and §8.1 shows that adding the class at this sample size did not remove the collision: the gap is the symptom, not a model-specific bug. The 100 GB patch budget requires H200-class hardware. Future work: (1) pseudolabelling the sacrum in VerSe's [Sekuboyina et al., 2021] 374 scans, whose scheme includes L6 but which have no pelvis, for a spinopelvic cohort of up to $1{,}527$ volumes; (2) two-stage integration with VERIDAH, our segmenter as its CT upstream.

Limitations. COLONOG is a US screening cohort (ages 45–75); generalization to emergency/pediatric populations is not evaluated. The six subtypes are functions of the source annotations' filenames and lumbar counts, not image reads. The Castellvi grades are a two-resident consensus (κ=0.74, n=33) read after peer review by co-authors who had seen a May 2026 grading of the same cases, with no senior read, no blinded re-read and no screening of the `normal` cohort for Type I; the source-dataset construction biases against morphologically mild Type I presentations (2 of 33, both in Table 1, neither held out), so Castellvi-stratified statements are scoped to the conspicuous regime. The held-out split was drawn by record, so 92 of its 167 patients have their other acquisition in training. Seven of the 33 LSTV patients have an LSTV-tagged held-out record (3 lumb, 1 sacr_count, 2 full sacralization, 1 ambiguous, 0 semi-sacralization, which is therefore unevaluated), and junction-DSC is scorable on one lumbarization and one full-sacralization case among them; fold-level evaluation pools five checkpoints over one 174-record test set (§8.1), and the Wilson intervals for strata of 3–4 cases start at 44–51%. The unmerged control (§8.1) uses a different split and is read from the per-epoch validation confusion block within its cross-validation folds rather than from a full-volume evaluation on the held-out split. TS is evaluated zero-shot, as a production system; its later `total_v3` task (with L6) postdates the benchmark and is not evaluated here. The sacrum comparison is not like-for-like: TotalSegmentator emits the S1 body as a separate class, which the benchmark's sacrum score counts as a miss, so 0.817 understates TotalSegmentator and the $+15.2$-point margin is an upper bound; the level-shift findings of §3, which concern lumbar labels, do not depend on it. The wrong-level framing is a mechanism, not a measured outcome: no reader study, planning simulation or workflow evidence is presented.

Implications. A production CT segmenter that emits a well-formed, caudally shifted prediction on a 4–30%-prevalence variant gives no error signal at evaluation time and, when consumed downstream, would reproduce the pattern of the wrong-level miscount. Any model claiming readiness for surgical planning should report per-subgroup DSC and L5/S1 confusion on a stratified LSTV cohort, with n.

Data and Code Availability

CTSpinoPelvic1K is archived on Zenodo under concept DOI 10.5281/zenodo.22139642 (current version v11, 10.5281/zenodo.22797014); §5 describes the release used; v6 holds a Croissant file with core and Responsible AI fields [Akhtar et al., 2024]. The CT volumes are mirrored at `https://huggingface.co/datasets/OpenSpineConsortium/CTSpinoPelvic1K`. The five fold checkpoints were trained on 3 May from a working tree at commit `50b209f` of a private development repository with the then-uncommitted v20 trainer, committed there on 5 May as `df10fea`; they are at `https://huggingface.co/OpenSpineConsortium/spinopelvic-seg-checkpoints`. A copy of that trainer, changed only in naming and anonymization, with the conversion and inference code, is at `https://github.com/OpenSpineConsortium/spinopelvic-seg` (commit `c105981`). The data-construction pipeline, the evaluation protocol and the TotalSegmentator benchmark script are at `https://github.com/Gregory-Schwing-MD-PhD/CTSpinoPelvic1K`. All are open without credentials; the April release's records and the evaluated split are not yet public (Appendix A).

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A Release used, records and the evaluated split

The April 2026 release and Zenodo v11. The segmentation experiments ran on the April 2026 release (§5). Its separate spine-only and pelvic-only records, for patients whose two source annotations lie on different acquisitions, are what the partial-annotation protocol (§5) trains on. The current archived version, v11, differs from it as follows: one record per patient (802 records), each separate-mode patient's missing pelvis completed on the spine's series by out-of-fold pseudolabelling; VerSe-native identifiers (L1–L6 = 20–25, sacrum 26, hips 30/31) with femora, per-level ribs, lumbar ribs and surgical hardware added as further classes; hip-laterality corrections on 22 records and hand corrections on five; both Castellvi reads and their consensus in the manifest; and the CT images supplied by TCIA series crosswalk and rebuild script, with a HuggingFace mirror carrying the volumes. The 802 patients and the 33 Castellvi-graded patients are the same in both.

Records and split. Table 7 gives the composition and the record sets the experiments used. The train, validation and test manifests are those of the release's 1 May 2026 revision, redrawn two days before training began; the converter sends the train and validation manifests to the cross-validation pool and holds out the test manifest. The five folds are the release's patient-level fold assignment copied onto the pool's records, so within the pool no patient spans two validation folds. Each validation-manifest record therefore trains four of the five fold models, and each fold's best checkpoint is selected on that fold's validation records, never on the test manifest.

Table 7: CTSPINOPELVIC1K composition (April 2026 release) and the record sets of the experiments.

Set	Patients	Records
Fused (spine + pelvis)	342	342
Separate-mode (spine record + pelvic record)	351	702
Spine-only / pelvic-only	89 / 20	89 / 20
All	802	1,153
with spine / pelvic labels	—	782 / 713
of the 33 LSTV patients	33	53
Train manifest	639	805
Validation manifest	170	174
Cross-validation pool (train + validation)	—	979
Test manifest (held out)	167	174

The test manifest is a record-level split stratified by LSTV class and match type. Of its 174 records (167 patients), 82 records from 75 patients are patient-disjoint from the pool: 52 fused and 16 single-mode patients (14 spine-only, 2 pelvic-only), and 7 separate-mode patients with both records held out. The other 92 are separate-mode records (52 pelvic, 40 spine) whose sibling—the patient's other acquisition, labelled only for the other region—is in the pool (81 from the train manifest, 11 from the validation manifest) and trains four of the five fold models. Among the 33 LSTV patients, 167, 672 and 125 are held out entirely; 401, 555, 123 and 4 have an LSTV-tagged held-out record and a pool record (tagged normal for 401, 555 and 123 by the refinement rule of §5, ambiguous for 4); 104 has its spine record (tagged sacr_count) in the pool and its pelvic record, tagged normal, held out. The pool siblings of 104 and 4 carry an LSTV tag and were therefore oversampled in the folds that trained on them. By record, the 33 LSTV patients' 53 records are 44 in the pool (28 LSTV-tagged) and 9 held out (7 LSTV-tagged). Table 3 counts patients: the Pool and Test columns overlap in one patient (4) by tag, and in five (401, 555, 123, 4, 104) by any record.

The Normal stratum. The Normal rows of Tables 4 and 5 hold 167 records from 161 patients: 165 records of 159 normal patients and the pelvic records of LSTV patients 104 (count-style sacralization) and 672 (lumbarization), tagged normal because they carry no spine labels. Removing those two records moves the Normal ensemble sacrum mean from 0.9719 to 0.9717, the merged hip from 0.9884 to 0.9882 and the mean foreground Dice from 0.9658 to 0.9655; spine classes and junction-DSC are unaffected.

What is public. Neither the April release's 1,153 records nor the evaluated split is at a location this paper cites. The Zenodo archive (v11) and the OpenSpineConsortium HuggingFace mirror hold the 802-record v11 layout, and the archive's `splits_5fold.json` is the patient-level fold assignment

over the 802 patients, with no test split. Its per-patient subtype map is not the six-way taxonomy of this paper: like the unmerged control's (Appendix C), it has no count-style class and tags one more patient lumbarization (lumbarization 15, sacralization 15, semi-sacralization 2, ambiguous 2, normal 768), so strata drawn from it differ from those of Tables 1 and 3. The six-way subtypes are the `lstv_phenotype` field of the v11 manifest (lumbarization 14, count-style sacralization 9, full sacralization 6, semi-sacralization 2, ambiguous 2, empty for the 769 normal patients). The anonymous HuggingFace dataset available during review had been re-uploaded on 18–20 May 2026 with a different record-level split, which shares 25 of the 174 held-out records. Until the April manifests and the held-out record list are published, the segmentation results cannot be reproduced exactly from public files.

B Castellvi grades before and after review

The reviewed version's Table 4 was a single grade column dated 6 May 2026, described there as a joint annotation by the two residents with no independent inter-rater scoring. An intermediate sheet of 23 August 2026 added second reads of five cases (175, 189, 215, 261, 267) and changed one grade (175, IIIb→Ib). The two full reads, complete by 4 September 2026, and their consensus are what this paper reports (Table 8). Reader 2's grades for those five cases equal the 23 August second reads, including 189 and 215, where reader 1 differs, so at least part of reader 2's grading predates September.

Table 8: Castellvi distribution over the 33 LSTV patients in each grade sheet.

Sheet	Ib	IIa	IIb	IIIa	IIIb	IV
May 2026 (reviewed Table 4)	1	2	4	3	19	4
August 2026 (five second reads)	2	2	4	3	18	4
Reader 1 (by 4 September 2026)	2	4	6	1	16	4
Reader 2 (by 4 September 2026)	1	2	6	3	17	4
Consensus (4 September 2026)	2	4	6	1	16	4

What changed. Five grades differ between the May sheet and the consensus: tokens 22 and 140 (IIIa→IIa), 123 and 125 (IIIb→IIb), and 175 (IIIb→Ib). In the four III→II changes the May sheet and reader 2 agreed on Type III and the consensus took reader 1's Type II; 175 is Ib in both reads. Type I therefore rose from 1 to 2 (175 and 215, both lumbarizations), IIIb fell from 19 (58%) to 16 (48%), and full sacralization went from all Type III/IV to IIa 1, IIb 2, IIIb 1, IV 2. All nine count-style sacralizations are IIIb in the May sheet and in both reads.

Agreement. The six disagreements (reader 1/reader 2) are 22 IIa/IIIa, 123 IIb/IIIb, 125 IIb/IIIb, 140 IIa/IIIa, 189 IIIb/IIb and 215 Ib/IIb. The consensus equals reader 1 on all 33 cases and reader 2 on 27. The May sheet agrees with reader 2 on 30 of 33 and with reader 1 on 28; the files do not record who filled it. A case bootstrap of the full-grade κ (patients resampled, B=10,000) gives 0.53–0.91, against the Fleiss–Cohen–Everitt interval 0.55–0.92 of §5. The per-reader files carry no timestamps, so whether the two reads were made independently of each other cannot be checked from them. The September sheet also records a count of non-rib-bearing vertebrae in a single column: the sheet's count of 6 (as in May) carries reader 1's annotation of 4 or 5 in 7 of 33 cases (the six full sacralizations and semi-sacralization 22); the count is not used in this paper.

C The unmerged control: split, loss and readings

Split and subtype map. The unmerged control (§8.1) was trained on 26–31 July 2026 on a later conversion of the same 1,153 records: another record-level 805/174/174 split with a 979-record cross-validation pool, fold train/validation sizes 765/214, 791/188, 798/181, 774/205 and 788/191. It shares 25 of the 174 records held out in this paper; 149 of those 174 lie in its pool, among them held-out LSTV records of patients 167 (its validation fold 3), 672 (fold 0), 123 (fold 2), 4 (fold 1) and 104 (fold 4). Its subtype map has no count-style class (count-style sacralization is merged into sacralization) and tags 15 records lumbarization, one more than this paper's taxonomy (token 500).

Loss and sampling. Cross-entropy weights were $[0.5, 1, 1, 1, 1, 1.5, 4.0, 2.0, 1, 1]$ (background, L1–L4, L5, L6, sacrum, hips), the merged model's weights plus L6 at 4.0. Patch-level L6 biasing was configured at probability 0.60 and forced an L6 patch in 0.98–1.41% of foreground-forced draws per fold; the merged model ran without patch-level biasing.

Table 9: Unmerged control per fold: validation lumbarization records (tokens), share of GT-L6 voxels predicted L5 (mean and range over the final 100 epochs; final epoch), and validation pseudo-Dice over the same epochs.

Fold	Lumbarization (val)	L6→L5, mean (range)	Final	L6 Dice	L5 Dice
0	500, 672, 737	83.0% (29.0–99.9)	99.9%	0.092	0.950
1	none	—	—	0.099	0.928
2	189, 261, 666	65.4% (26.0–99.5)	93.3%	0.047	0.936
3	167, 267, 344	66.3% (17.9–98.2)	64.7%	0.134	0.951
4	175, 215, 587	96.8% (64.3–100)	99.8%	0.014	0.907

Fold 4's validation lumbarizations include both Type I patients (175 and 215). Fold 1's confusion block reports no lumbarization case in any of its 500 epochs.

The figure quoted during review. A range of 84.7–99.4% was quoted on 27 July 2026, about 17:00. At that time only folds 0 and 4 were running, at epochs 279 and 277 of 500; folds 1–3 started on 28 July. The logs admit one moment at which the latest single-epoch readings of folds 0 and 4 were 84.7% and 99.4%: 27 July, 14:02–14:06 (epochs 257 and 255). The figures in §8.1 are from the completed runs, under the rule stated there.

D Training and hardware details

Network. nnU-Net v2 ResEnc-M planner, `3d_fullres`, `nnUNetResEncUNetPlans_100G`: a 7-stage residual encoder with n_{features}=$[32, 64, 128, 256, 320, 320, 320]$ and n_{blocks}=$[1, 3, 4, 6, 6, 6, 6]$, 3^3 kernels, 2^3 strides, instance normalization and deep supervision at 5 levels; a 9-channel output (background and eight foreground classes), with the ignore label (9) masked from the loss. In the class-reweighted DC_and_CE loss, $\mathbf{w}$=$[0.5, 1, 1, 1, 1, 1.5, 2.0, 1, 1]$, the 1.5 on last_lumbar targets the lumbosacral boundary and the 2.0 on sacrum the junction the Castellvi grade describes.

Hardware. Training ran on H200 NVL (140 GB HBM3e) with 24 CPU workers. The preprocessed data were unpacked once per job to a 1.7 TB local-NVMe cache, replacing network-file reads that throttled at 21 MB/s when five folds read at once.

E Per-class n and intervals for the stratified results

Table 10 gives, for each row of Table 1, the number of patients scorable for each class and a 95% case-bootstrap interval for the mean: patients resampled with replacement within the subgroup, B=10,000, percentile, seed 20260926. Where three or fewer patients are scorable the per-patient values are listed instead. A class is unscored for a patient when none of its records carries that region's labels. The Normal−Any-LSTV junction drop of 29.0 points has a two-sample case-bootstrap interval of 9.9–47.4 points; under the stricter rule of §3 (5 LSTV patients) the drop is 35.5 points (15.1–55.5). Hips are omitted: no row differs from Normal by more than 0.01.

Our model, per case. No population interval is possible for our held-out LSTV strata, which hold one to three records per class. Table 5 gives the mean of the five fold checkpoints; the per-record values below are the five-fold ensemble's, which is what the per-case evaluation file holds. On the three held-out lumbarization records (167, 401_r1, 672), L1 is .805, 1.000 and .797; L2 .492, 1.000 and .403; L3 .175, .988 and .006; L4 .0003, .675 and .0003; and last_lumbar .645, .725 and .666 (ensemble means .867, .632, .389, .225 and .678, against the checkpoint means .872, .611, .377, .213 and .698 of Table 5). Two of the three (167 and 672) shift through the upper lumbars; 401_r1 does not.

Table 10: TS zero-shot: 95% case-bootstrap interval of each Table 1 mean, with the number of scorable patients (n) below it; per-patient values where three or fewer are scorable.

Subgroup	L1	L2	L3	L4	L5	Sac	Jxn DSC
Normal	.909–.931	.906–.929	.920–.937	.928–.944	.931–.945	.816–.817	.820–.828
n	750	750	750	750	750	683	334
Any LSTV	.535–.820	.459–.763	.385–.660	.226–.513	.175–.458	.810–.833	.348–.722
n	31	31	31	31	31	29	7
Lumbarization	.740–.953	.710–.943	.545–.817	.184–.602	.099–.507	.816–.854	.221, .234, .618
n	14	14	14	14	14	11	3
Sacr. (count)	.011–.532	.000–.296	.000–.163	.000–.001	.000–.000	.805–.822	.467, .927
n	9	9	9	9	9	8	2
Semi-sacralization	.951	.951	.951	.955	.956	.821, .828	none
n	1	1	1	1	1	2	0
Sacr. (full)	.339–.952	.297–.953	.317–.953	.371–.954	.500–.953	.801–.851	.433, .838
n	5	5	5	5	5	6	2
Ambiguous	.951, .952	.949, .953	.863, .901	.649, .707	.336, .413	.730, .813	none
n	2	2	2	2	2	2	0

F Background: transitional vertebrae, Castellvi grading and VERIDAH

Enumeration and transition. The spine usually has seven cervical, twelve thoracic and five lumbar vertebrae. An enumeration anomaly gives eleven or thirteen thoracic or four or six lumbar vertebrae; a transitional vertebra is a border segment with features of both neighbouring regions [Möller et al., 2026, Konin and Walz, 2010]. The two co-occur: a lumbosacral transitional vertebra (LSTV) is associated with an anomalous total count [Carrino et al., 2011]. An LSTV is either sacralization of the lowest lumbar segment, from a broadened transverse process to complete fusion with the sacrum, or lumbarization of the uppermost sacral segment, which may articulate rather than fuse and may show lumbar-type facets or a fully sized S1–S2 disc [Konin and Walz, 2010]. The same words also name counts: Nagata et al. [2025] define sacralization as 23 presacral vertebrae and lumbarization as 25. Rib conventions differ as well: VERIDAH calls any vertebra with a true rib thoracic, whatever the rib's length [Möller et al., 2026]; VerSe'20 calls it thoracic only if one rib exceeds 38 mm, and lumbar if both are shorter [Liebl et al., 2021]. The reference standard for numbering remains a count from C2 on whole-spine imaging [Lian et al., 2018]. Reported LSTV prevalence is 4–30% [Konin and Walz, 2010], 21.1% of 1,468 lumbar MRIs [Hanhivaara et al., 2022] and 16.3% of 551 whole-spine CTs by count [Nagata et al., 2025]. It varies with the population, from 8.1% to 16.9% across three clinic groups in one series, which did not count Castellvi type I as transitional [Sekharappa et al., 2014].

The six subtypes in these terms. The taxonomy of §5 inherits its sources' conventions. CTSpine1K numbers C1–L6 as 1–25 and attributes an L6 to sacral lumbarization and a missing L5 to lumbar sacralization [Deng et al., 2021]; `lumb` and `sacr_count` are these counts. A count does not place the anomaly: six lumbar vertebrae can follow eleven thoracic ones, and four can follow a T13 (Möller et al., 2026, Figs. 1D and 5), and no scan here, all CT colonography of the abdomen and pelvis, contains C2 [Schwing et al., 2026]. `sacralization` and `semisacralization` rest on CTPelvic1K's undocumented filename tag. The April release has no thoracic, T13 or rib class, so the thoracolumbar border cannot be checked on it; the current release (v11) adds thoracic levels, ribs, lumbar ribs and a T13 class [Schwing et al., 2026].

Castellvi classification. Castellvi et al. [1984] typed the transitional segment mainly by the morphology of its transverse process. As restated by Konin and Walz [2010], with *a* marking a unilateral and *b* a bilateral finding, type I (Ia, Ib) is a dysplastic transverse process at least 19 mm in craniocaudal width. Type II (IIa, IIb) is an enlarged transverse process forming a diarthrodial joint with the sacrum. Type III (IIIa, IIIb) is complete osseous fusion of the transverse process to the sacrum. Type IV is type II on one side and type III on the other. The grade "does not provide information relevant to accurate enumeration of the involved segment" [Konin and Walz, 2010, p. 1778], which is why §5 treats it as orthogonal to the count; type IIIb occurs here at lumbar counts of four, five and six (Table 3). VerSe'20 classified only types II–IV, lists each subject's variants, and left vertebrae partly

fused to the sacrum (III, IV) unsegmented [Liebl et al., 2021]; VERIDAH's reference labels call any Castellvi-bearing vertebra a sacralized lumbar one, and it outputs no grade [Möller et al., 2026].

VERIDAH and SPINEPS. VERIDAH [Möller et al., 2026] labels vertebrae whose positions are given (the paper takes them from instance segmentations). A 3-D DenseNet169 classifies a crop centred on each vertebra with four heads: vertebra label (no T13 or L6 class), region, transition (last cervical, first thoracic, last thoracic, first lumbar, last lumbar, or none) and visibility. For training, T13 and L6 are relabelled T12 and L5, so an extra vertebra is a repeated T12 or L5 that only the transition head separates, and anomalous cases are drawn four times per epoch. A sequence predictor combines the heads into a path cost weighted by each vertebra's visibility and minimises it by dynamic programming under hard constraints: consecutive, spatially ordered labels; 11–13 thoracic and 4–6 lumbar; at most one repeated T12 and one repeated L5, renamed T13 and L6. On 365 in-house test CTs enriched for anomalies it labelled every vertebra correctly in 99.18% of subjects, against 77.81% for the method of Meng et al. [2022], and recalled 97.22% of lumbar enumeration anomalies; in its ablation one label head without the predictor recalls 30.56%, with it 63.89%, and four heads without it 30.56%. SPINEPS [Möller et al., 2024] is a semantic-then-instance segmentation of T2-weighted MRI whose instances are counted top to bottom "regardless of the field of view or the presence of enumeration abnormalities"; VERIDAH supplies the labelling step and ships with its code [Möller et al., 2026]. SPINEPS 2.1.2 (the current release) runs CT models before the VERIDAH labeller and writes VerSe ids, L6 as 25 and T13 as 28 [Möller, 2026]; the peer-reviewed SPINEPS paper covers MRI only.

Why a missing class shifts labels. A segmenter gives each voxel one label from a fixed vocabulary. VerSe numbers L1–L6 as 20–25 and T13 as 28 [VerSe organisers, 2026], and CTSpine1K and the April release have an L6 class; the `total` task of TotalSegmentator (version 2.4.0 was audited) has C1–L5, the sacrum and S1 but no L6 or T13 [Wasserthal et al., 2023, Wasserthal, 2026].[2] Given six lumbar-type segments and five lumbar labels, a model must reuse a label or give a segment a thoracic, sacral or no label; on lumbarized spines TotalSegmentator gives the L6 the L5 label in two of three scorable junctions, and its lumbar Dice falls level by level toward the sacrum (§3). An L6 output label is necessary but not sufficient, and it need not be a trained class. CTSpine1K's own nnU-Net baseline scored an L6 Dice of almost zero [Deng et al., 2021]; our unmerged control, trained with an L6 class, still labels most ground-truth L6 voxels in its validation patches L5 (§8.1); and VERIDAH as published has no L6 class, yet names an L6 as a repeated L5 that its transition head and sequence predictor place [Möller et al., 2026].

G How the April 2026 release was built

Figure 2 traces the April 2026 release (§5) from its three sources to the records the experiments used. It is adapted from the construction figure of the dataset article [Schwing et al., 2026], which describes the later v11 release; the steps only v11 has are left out and are listed in Appendix A. The placement step counts 804 patients, 352 of them separate and 90 spine-only. The orientation review excluded one separate patient, and one placed spine-only patient is absent from the export for a reason the export did not record, so the release holds 351 separate and 89 spine-only patients. A front-to-back mirror is not a rotation. In the two of the nine checked, tokens 75 and 480, the released CT shows the liver on the left. The six hip-label swaps match the labels to the side of the image, which for 480 is the mirror of the patient's; the other seven were not rendered.

H Comparison with other CT collections

Table 11 compares CTSpinoPelvic1K with the CT collections that label anatomy at the lumbosacral junction; each cell was checked against the collection's own paper or data page. VerSe comes closest in intent: it labels L6 and T13, lists anatomical variants per subject and grades Castellvi II–IV, but it segments neither the sacrum nor a transitional vertebra partially fused to it, so its type III and IV cases

[2] Since version 2.15.0 (1 July 2026) the TotalSegmentator package has also defined a `total_v3` task whose class map puts `vertebrae_L6` where `total` has `vertebrae_S1`; its weights were posted as a GitHub pre-release on 7 September 2026, and neither the changelog nor the README describes it [Wasserthal, 2026]. It postdates the benchmark of §3. It was not evaluated here.

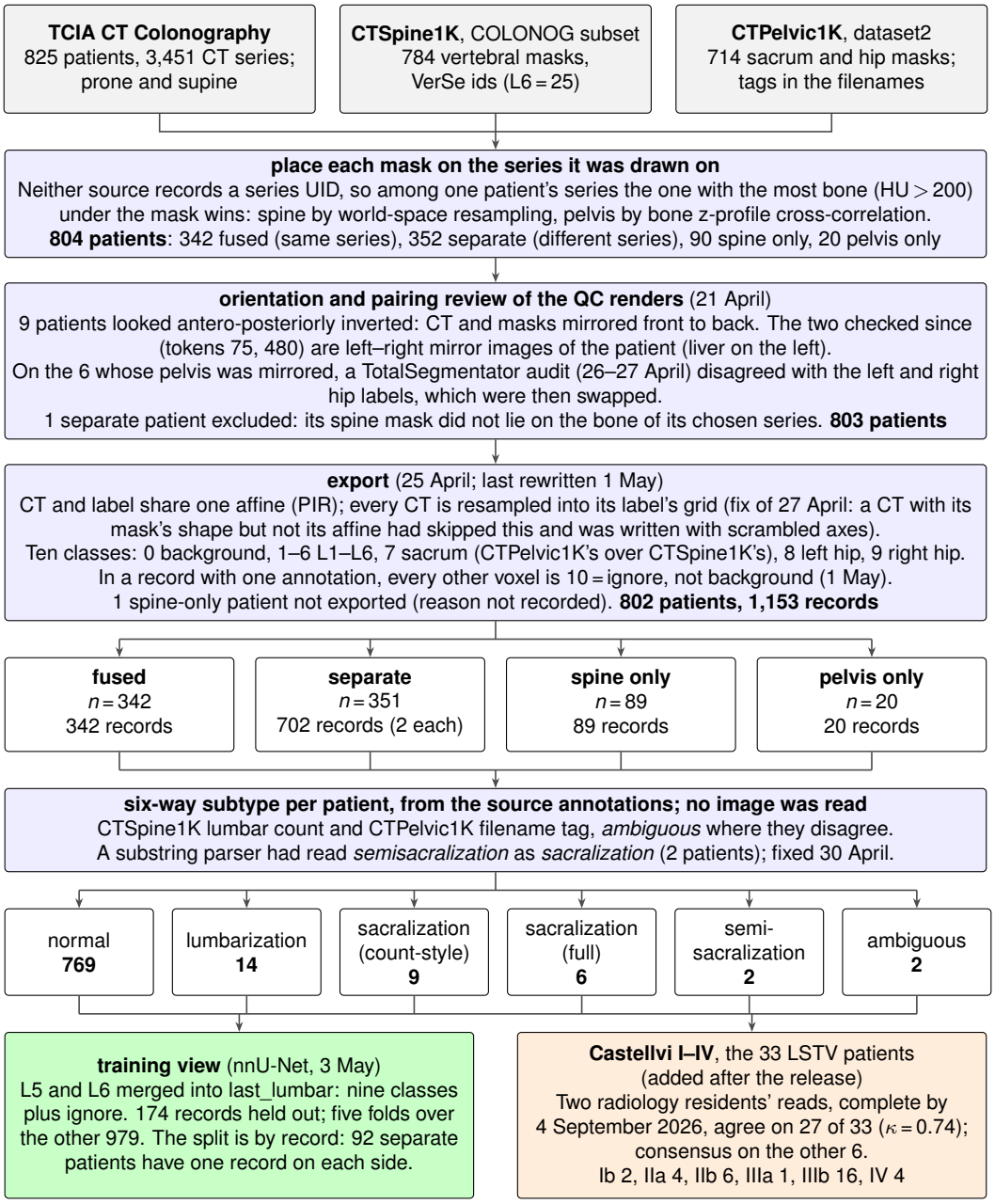


Figure 2: Construction of the April 2026 release of CTSPINOPELVIC1K, on which every segmentation experiment here was run: 802 patients, 1,153 records; *n* counts patients. Grey, sources; blue, the April pipeline; green, the training view; orange, added after the release. Adapted from Fig. 1 of the dataset article [Schwing et al., 2026], which describes a later release (v11).

cannot be scored at L5/S1. CTSpine1K labels L6 but no pelvis, and its nnU-Net baseline already scored L6 at a Dice of 0 on all three of its test sets, with shifted predictions on lumbarization and sacralization cases (its Table 2 and Fig. 4) [Deng et al., 2021], which is consistent with the unmerged control of §8.1. Among these collections, only CTSPINOPELVIC1K puts numbered lumbar levels including L6, the sacrum and both hip bones on the same patients, with a per-patient LSTV label and, for its 33 LSTV patients, a Castellvi grade.

Table 11: CT collections labelling the lumbosacral junction. ✓ present; (✓) partial; – absent; letters refer to the notes. Patients in parentheses where reported and different from scans. Pelvis: S sacrum, H hip bones, F femora. Unified: spine and pelvis in the same patients. Adapted from Table I of the dataset article [Schwing et al., 2026], which describes a later release.

Collection	**Scans** (patients)	**Levels** labelled	**Pelvis**	**Unified** cohort	**LSTV** labels	**Six-way** subtype	**Castellvi** grade	**Public** data
VerSe	374 (355)	C1–L6, T13[a]	–	–	(✓)[b]	–	(✓)[c]	✓
CTSpine1K	1,005	C1–L6[d]	–	–	(✓)[d]	–	–	✓
CTPelvic1K	1,184	lumbar, 1 class[e]	S, H	(✓)[e]	(✓)[f]	–	–	✓
TotalSegmentator	1,204	C1–L5[g]	S, H, F	(✓)[g]	–	–	–	✓
VERIDAH CT cohort	1,536	C1–L6, T13[h]	–[h]	–	(✓)[h]	–	–[h]	–
CTSPINOPELVIC1K	1,153 (802)[i]	L1–L6	S, H	✓[i]	✓[j]	✓[j]	✓[k]	✓[i]

[a]Identifiers 1–25 (C1–L6) and 28 (T13); sacrum and coccyx not labelled [VerSe organisers, 2026]; 374 scans of 355 patients [Sekuboyina et al., 2021]. [b]VerSe'20 was assembled partly from reports of anatomical variants and lists them per subject; 85 of its cases have six lumbar vertebrae [Liebl et al., 2021]. [c]Castellvi II–IV graded in VerSe'20 (IIa 35, IIb 25, IIIa 14, IIIb 28, IV 7), types 0 and I pooled; the sacrum and vertebrae partially fused to it (III, IV) are not segmented [Liebl et al., 2021]. [d]Identifiers 1–25 (C1–L6); 784 of the 1,005 volumes are COLONOG. Its Table 1 counts 17 lumbarization and 11 sacralization cases, ungraded; the authors list their IDs with the data [Deng et al., 2021]. [e]Four classes: lumbar spine (one class), sacrum, left and right hip, on the same scan, cropped to the pelvic region; patient count not reported [Liu et al., 2021]. [f]Undocumented tags in ten of the 713 COLONOG mask filenames on Zenodo: sacralization 7, semisacralization 2, hard_sacralization 1 [Liu et al., 2020, Schwing et al., 2026]. [g]The cited paper's dataset (v1, 104 classes) [Wasserthal et al., 2023, Wasserthal, 2022]; v2.0.1 has 1,228 scans and 117 classes and adds S1 [Wasserthal, 2023, 2026]; neither has L6 (the software's later `total_v3` task has an L6 class, not evaluated here). Every structure in the field of view is labelled; how many scans hold both lumbar spine and pelvis is not reported. [h]In-house cohort, patient count not reported; reference labels include T13 and L6, merged into T12 and L5 for training; enumeration anomalies are labelled by count, and a Castellvi transitional vertebra is labelled sacralized lumbar whatever its type; lumbar and sacral vertebrae are distinguished, but no sacral or pelvic segmentation is described; code and weights public, no image release described [Möller et al., 2026]. [i]Records of the April 2026 release used here (Appendix A). Spine and pelvis for 693 of 802 patients: on one series for 342, on two acquisitions of the same patient (two records) for 351; 89 spine-only, 20 pelvis-only. Public as v11 (the same 802 patients, one record each); the April records and evaluated split are not yet at a cited location. [j]For all 802 patients (33 LSTV), from the source masks' lumbar counts and filename tags, not image reads (§5); the six subtypes are listed per patient in the v11 manifest (`lstv_phenotype`). [k]Types I–IV for all 33 LSTV patients: a two-resident consensus completed in September 2026, after the April release, and distributed from Zenodo v8 onward (§5, Appendix B); normal patients not screened for type I. Licences: VerSe CC BY-SA 4.0; CTSpine1K CC BY-NC-SA 4.0; CTPelvic1K CC BY-NC-SA (paper) or CC BY 4.0 (Zenodo record); TotalSegmentator data CC BY 4.0; CTSPINOPELVIC1K CC BY-NC-SA 4.0 on Zenodo, the archive of record, and CC BY-NC 4.0 on the dataset card of its HuggingFace mirror (checked 27 September 2026).

How the benchmark will be used. Its first use is as a stratified test: a CT segmenter or vertebra labeller is scored per LSTV subtype with junction-DSC and voxel confusion at L5/S1, each stratum reported with its n (§6), which is what a model claiming readiness for surgical planning should show. Adding an L6 class did not remove the L5/L6 collision at this sample size (§8.1), and the signal that separates a last lumbar vertebra from its neighbour is a count that one patch does not reliably contain, so the benchmark is meant as the test set for counting-aware and two-stage "segment, then number" designs, such as a segmenter feeding VERIDAH's sequence predictor (§9). Its LSTV strata are small (33 patients, 7 with an LSTV-tagged held-out record; Wilson intervals for strata of 3–4 cases start at 44–51%, §9), so it exposes a failure mode and its direction rather than ranking methods that differ by a few points.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: Each of the four abstract contributions is supported by a corresponding section: the TS level-shift audit (§3, Table 1, with the case-study mechanism in §4), CTSPINOPELVIC1K construction, the 6-way LSTV taxonomy and the origin of its labels (§5), the LSTV-stratified evaluation protocol (§6), the spinopelvic segmenter and SOTA-on-reproducible-models comparison (§7, Tables 4–5), the same-data unmerged control (§8.1) and the residual-collision experiment pointing to VERIDAH's sequence predictor (§8.1, Table 6). Quantitative abstract values (TS junction-DSC $0.824 \rightarrow 0.534$ on Any-LSTV, sacrum $0.817 \rightarrow 0.969$, κ=0.74) are reported in the cited sections with their n. Two different sevens appear: TS's junction is scorable on 7 LSTV patients across the whole cohort (§3), and 7 held-out records carry an LSTV subtype (§7); the two sets share two patients, and our model's junction-DSC rests on one held-out lumbarization case. The unmerged control uses the same records but not the same split (§8.1). The held-out split was drawn by record, and the paper says so (§7). The wrong-level statement is scoped in the abstract, §1 and §4 as a plausible automated risk mechanism, not a measured clinical outcome.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: §9 contains a dedicated "Limitations" paragraph: COLONOG is a US screening cohort (ages 45–75) so generalization to emergency or pediatric populations is not evaluated; the six LSTV subtypes are functions of the source annotations' filenames and lumbar counts, not image reads (§5); Castellvi typing of all 33 LSTV cases is a two-resident consensus (κ=0.74, n=33, six disagreements) completed by 4 September 2026, after peer review (reader 2's grades for five cases are second reads recorded on 23 August), by co-authors who had seen a May 2026 grading of the same cases (five grades changed against the reviewed table; Appendix B), with no senior read, no blinded re-read and no screening of the `normal` cohort for Type I, so Castellvi-stratified statements are scoped to the conspicuous regime (31 of 33 cases are Castellvi II–IV; the two Type I patients, 175 and 215, both lumbarizations and 215 the case study, are in Table 1's lumbarization and Any-LSTV rows and fold 4 of the unmerged control, and the held-out split holds neither); the held-out split was drawn by record, so 92 of its 167 patients have their other acquisition in training; 7 LSTV patients have an LSTV-tagged held-out record (3 lumb, 1 sacr_count, 2 full sacralization, 1 ambiguous, 0 semi-sacralization, which is unevaluated), junction-DSC is scorable on one lumbarization and one full-sacralization case among them, and the Wilson intervals for such strata are wide; the unmerged control uses a different split and is read from per-epoch validation confusion within its folds rather than a full-volume held-out evaluation; TS is evaluated zero-shot; and the wrong-level framing is a mechanism, not a measured outcome (no reader study or workflow evidence).

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DC+CE loss with explicit weight vector, optimizer schedule, 500 epochs, two-tier LSTV emphasis, and 5-fold CV grouped by patient and stratified by LSTV subtype over a 979-record pool, with a 174-record held-out split drawn by record (§7, Appendix A). The construction pipeline, trainer source, evaluation code, and trained 5-fold checkpoints are released; see "Data and Code Availability". Neither the April release's records nor the evaluated record-level split is yet at a public location the paper cites (Appendix A).

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v20 trainer, committed there on 5 May as df10fea), and the data-construction pipeline, evaluation protocol and TotalSegmentator benchmark script (GitHub, Gregory-Schwing-MD-PhD/CTSpinoPelvic1K) are public without credentialed access. What is not yet public at a location the paper cites: the April release's 1,153 records (the archive and the HuggingFace mirror hold the 802-record v11 layout) and the record-level split evaluated here (the archive's frozen split is a patient-level fold assignment with no test split; Appendix A). See "Data and Code Availability".

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   Justification: Our segmenter is trained with 5-fold CV grouped by patient within a 979-record pool and evaluated on the 174-record held-out split, which was drawn by record

(§7); per-subgroup DSC are reported as means $\pm$ one standard deviation across the five fold checkpoints evaluated on the same held-out split (Table 5), which captures checkpoint variability, not population variability—the captions say so, and cells that rest on a single record are marked. Per-case values for the three held-out lumbarization records are given for every lumbar class (Appendix E; last_lumbar also in §8); the full-sacralization row is one record for its spine classes and two for its pelvic classes. Δ against TS in Table 5 is paired on the same held-out records; the sacrum gain in §8 is labelled unpaired. Proportions on small strata carry Wilson 95% intervals (closed form; e.g. 3/3 last_lumbar emission on lumbarization: 44–100%; 4/4 on the held-out LSTV records with last_lumbar GT: 51–100%; 10/14 lumbarization cases with TS L5 DSC below 0.5: 45–88%). Inter-rater agreement on the Castellvi grades is Cohen's κ with Fleiss–Cohen–Everitt asymptotic 95% intervals (full grade 0.74, 0.55–0.92; type only 0.70, 0.48–0.92; n=33); a case bootstrap (B=10,000) gives 0.53–0.91 for the full grade. TS (Table 1) is evaluated zero-shot once per case; we report per-case means with n per subgroup, the n of scorable junctions, per-case values for the junction strata, and, in Appendix E, the per-class n with a 95% case-bootstrap interval for every subgroup mean over four or more patients (patients resampled within the subgroup, B=10,000, percentile), e.g. lumbarization L4 0.388 (0.18–0.60) and L5 0.291 (0.10–0.51), n=14; the 29-point Any-LSTV junction drop has a two-sample case-bootstrap interval of 10–47 points (§3). Our model's LSTV strata hold one to three records per class, too few for an interval, so their per-case values are given instead. No normality assumption is made; the unmerged control (§8.1) reports per-fold means over the final 100 epochs with the within-fold epoch range (per fold in Appendix C).

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   Answer: [Yes]

   Justification: §7 (Hardware) and Appendix D specify H200 NVL (140 GB HBM3e), 24 CPU workers, a 1.7 TB local-NVMe preunpack cache replacing throttled NFS reads, per-epoch $\approx$ 400–700 s, and per-fold $\approx$ 3 days for 500 epochs (5 folds $\times$ 3 days $\approx$ 15 GPU-days). The same paragraph documents that the 100 GB GPU-memory budget is the smallest configuration fitting both lumbar spine and pelvis at full spacing; smaller budgets force coarser patches or spacing. The unmerged control (§8.1) is a second 5-fold, 500-epoch run of the same network (trained 26–31 July 2026). The TS benchmark is one zero-shot

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